

Machine Learning Lab 6 : Convolutional Neural Networks (CNN)

NAME:-MOHAMMED BILAL

SECTION:-F

SRN:-PES2UG23CS344

DATE:-20-11-2025

1. Introduction

The objective of this lab was to build, train, and evaluate a Convolutional Neural Network (CNN) that can classify images of hand gestures—rock, paper, and scissors.

Using a dataset containing examples of each gesture, we designed a custom CNN model in PyTorch, trained it on the dataset, and then assessed how well it could recognize the gestures in unseen test images.

2. Model Architecture

The CNN model is made up of **three convolutional blocks** followed by a **fully connected classifier**.

◆ Convolutional Blocks

Each block contains the following layers:

- A **Conv2d** layer with a 3×3 kernel and padding of 1
- **ReLU** activation
- A **MaxPool2d** layer to reduce the spatial size by half

Block Structure:

Block	Input Channels	Output Channels	Kernel	Padding	Pooling
1	3	16	3×3	1	MaxPool2d (2×2)

2	16	32	3×3	1	MaxPool2d (2×2)
3	32	64	3×3	1	MaxPool2d (2×2)

After all three pooling operations, an input image of **128×128** is compressed to a feature map of **16×16** with **64 channels**.

◆ Fully Connected Classifier

The classifier contains:

- **Flatten layer:** Converts the 64×16×16 tensor into 16,384 features
 - **Linear Layer:** 16,384 → 256
 - **ReLU activation**
 - **Dropout (p = 0.3)**
 - **Final Linear Layer:** 256 → 3 (one for each class: rock, paper, scissors)
-

3. Training and Performance

Hyperparameters Used

- **Optimizer:** Adam
- **Loss Function:** CrossEntropyLoss
- **Learning Rate:** 0.001
- **Epochs:** 10
- **Batch Size:** 32

Final Test Accuracy

The trained model achieved a test accuracy of **68.17%**.

4. Conclusion and Analysis

The CNN model performed reasonably well on the rock–paper–scissors classification task, achieving a test accuracy of **68.17%**. Despite being a simple architecture, it demonstrated how effective convolutional layers are for image-based prediction tasks.

Challenges Faced

- Properly normalizing and preprocessing the images
- Preventing overfitting due to a relatively small dataset

Suggestions for Future Improvement

- Increasing model depth by adding more convolutional layers
- Applying data augmentation to boost generalization
- Using training strategies like early stopping or learning rate scheduling